

Question	Answer	Mark
1(a)(i)	zero error or wrongly calibrated scale	B1
1(a)(ii)	reading scale from different angles or wrongly interpolating between scale readings/divisions	B1
1(b)(i)	$P = V^2 / R$ or $P = VI$ <u>and</u> $V = IR$	C1
	$P = 5.0^2 / 125$ or 5.0×0.04 or $(0.04)^2 \times 125$ $= 0.20 \text{ W}$	A1
1(b)(ii)	$\%V = 2\%$ or $\Delta V / V = 0.02$	C1
	$\%P = (2 \times 2\%) + 3\%$ or $\%P = (2 \times 0.02 + 0.03) \times 100$ $= 7\%$	A1
1(b)(iii)	absolute uncertainty in $P = (7 / 100) \times 0.20$ $= 0.014$	C1
	power = $0.20 \pm 0.01 \text{ W}$ or $(2.0 \pm 0.1) \times 10^{-1} \text{ W}$	A1

